

ASSIGNMENT : Logistic Regression (Loan Approval Prediction)

Submission Requirements

Students must submit:

Google Colab Notebook

All code

Exploratory Data Analysis

Model evaluation metrics

Regression equation

Interpretation of coefficients

Problem Statement

You are working as a Data Analyst in a financial institution.

The company wants to build a model to predict whether a loan application will be approved(1) or not approved (0) based on customer details.

You are given a dataset:

Dataset Link : [Loan Approval Prediction](#)

Your task is to perform Exploratory Data Analysis (EDA), build a Logistic Regression model, evaluate the model, and interpret the results.

Q1. Data Loading & Understanding

- a) Load the dataset into Python.
- b) Display the first 5 rows.
- c) Check the shape of the dataset.
- d) Display the data types of each column.

SOL :

- a) Load the dataset into Python.

Import the required libraries and load the dataset into a Pandas DataFrame using the appropriate file path.

- b) Display the first five rows.

Use the dataset preview function to understand the structure of the data and examine how the variables are organized.

c) Check the shape of the dataset.

Determine the total number of rows and columns present in the dataset. This helps in understanding the size of the dataset and planning further analysis.

d) Display the data types of each column.

Identify whether each variable is numerical or categorical. Understanding data types is essential for preprocessing and selecting appropriate analytical techniques.

Q2. Data Cleaning

a) Check for missing values in the dataset.

b) Handle missing values appropriately.

c) Identify categorical columns present in the dataset.

SOL :

a) Check for missing values in the dataset.

Examine each column to identify the presence and number of missing values. Missing data can negatively affect model performance if not handled properly.

b) Handle missing values appropriately.

Apply suitable imputation techniques based on the nature of the variable:

- Replace missing values in numerical columns using statistical measures such as the mean or median.
- Replace missing values in categorical columns using the mode (most frequent category).

Document the approach used and justify why it is appropriate.

c) Identify categorical columns present in the dataset.

List all variables that contain non-numerical information. These variables will later require encoding before model development.

Q3. Exploratory Data Analysis (EDA)

a) Plot the distribution of the target variable.

b) Analyze the relationship between one numerical feature and the target variable.

c) Analyze the relationship between Years of Employment and the target variable.

SOL : a) Plot the distribution of the target variable.

Visualize the frequency of approved and non-approved loan applications using an appropriate chart, such as a count plot or bar chart.

Objective:

To understand whether the dataset is balanced or imbalanced with respect to the target variable.

b) Analyze the relationship between one numerical feature and the target variable.

Select an important numerical variable (for example, **Applicant Income**) and compare its distribution across loan approval categories using suitable visualizations such as:

- Box plots
- Violin plots
- Histograms grouped by target category

Objective:

To determine whether the selected numerical feature influences loan approval decisions.

c) Analyze the relationship between Years of Employment and the target variable.

Investigate how the number of years an applicant has been employed relates to loan approval status.

Possible approaches include:

- Box plots grouped by loan approval status
- Average years of employment comparison
- Distribution analysis

Objective:

To assess whether employment stability contributes to the likelihood of loan approval.

Q4. Outlier Detection and Treatment

a) Detect outliers in numerical columns using the IQR method.

b) Treat outliers using capping techniques

Sol : a) Detect outliers in numerical columns using the IQR method.

For each numerical feature:

1. Calculate the first quartile (Q1).
2. Calculate the third quartile (Q3).
3. Compute the Interquartile Range (IQR):

$$IQR = Q3 - Q1$$

4. Determine the lower and upper boundaries:

$$Lower\ Limit = Q1 - 1.5 \times IQR$$

$$Upper\ Limit = Q3 + 1.5 \times IQR$$

5. Identify observations falling outside these boundaries as potential outliers.

b) Treat outliers using capping techniques.

Apply capping (Winsorization) to limit extreme values.

- Values below the lower limit should be replaced with the lower boundary.
- Values above the upper limit should be replaced with the upper boundary.

Objective:

To reduce the impact of extreme observations while preserving all records in the dataset.

Q5. Convert target variables into numerical format suitable for model building and drop unnecessary columns.

Sol : a) Convert the target variable into numerical format suitable for model building.

Transform the loan approval status into binary values if necessary.

For example:

- Approved → 1
- Not Approved → 0

Ensure that the target variable is compatible with Logistic Regression requirements.

b) Drop unnecessary columns.

Remove variables that do not contribute to prediction, such as:

- Unique identifiers (e.g., Loan ID)
- Duplicate information
- Columns deemed irrelevant after exploratory analysis

Objective:

To improve model efficiency and reduce noise in the dataset.

Q6. Feature Selection and Data Splitting

- a) Separate independent variables (X) and dependent variable (y).
- b) Split the dataset into training and testing sets.

Sol: a) Separate independent variables (X) and dependent variable (y)

The dataset should be divided into **independent variables (features)** and the **dependent variable (target)**.

- **Independent Variables (X):** All predictor variables that may influence the loan approval decision, such as applicant income, loan amount, credit history, years of employment, education, marital status, etc.
- **Dependent Variable (y):** The **Loan Status** column, which indicates whether the loan application was approved (**1**) or not approved (**0**).

Objective:

To identify the input variables that will be used to train the Logistic Regression model and the target variable that the model aims to predict.

- b) Split the dataset into training and testing sets

After separating **X** and **y**, divide the dataset into **training** and **testing** subsets using an **80:20 ratio**.

- **Training Set (80%)** – Used to train the Logistic Regression model and learn patterns from historical data.
- **Testing Set (20%)** – Used to evaluate the model's performance on unseen data and assess its generalization ability.

A fixed **random state** should be specified to ensure that the results are reproducible.

Objective:

To build the model using one portion of the data while reserving another portion for unbiased performance evaluation. Proper data splitting helps prevent overfitting and provides a realistic estimate of how the model will perform on new loan applications.

Q7. Apply feature scaling to the dataset using StandardScaler.

Sol : Apply **StandardScaler** to standardize the numerical features in the dataset.

Standardization transforms the data so that each feature has:

- Mean = 0

- Standard Deviation = 1

The scaler should be:

1. **Fitted on the training data only.**
2. **Applied to both training and testing datasets** using the parameters obtained from the training data.

Objective:

To ensure that variables measured on different scales contribute equally to the Logistic Regression model and improve model performance.

Q8. Logistic Regression Model Building

- a) Train a Logistic Regression model.
- b) Predict the output for test data.

SOL : a) Train a Logistic Regression model.

Use the scaled training dataset to fit a **Logistic Regression classifier**. The model will learn the relationship between applicant characteristics and the likelihood of loan approval.

Objective:

To develop a predictive model capable of classifying loan applications as approved or not approved.

- b) Predict the output for test data.

Use the trained model to generate predictions for the testing dataset.

The model should provide:

- **Predicted class labels (0 or 1)** indicating loan approval status.
- **Predicted probabilities** representing the likelihood of loan approval.

Objective:

To assess how effectively the model predicts outcomes for previously unseen data.

Q9. Model Evaluation – Confusion Matrix

- a) Generate the confusion matrix.
- b) Interpret the results.

SOL: a) Generate the confusion matrix.

Construct a **Confusion Matrix** to compare the actual loan approval outcomes with the model's predicted outcomes.

The matrix consists of four components:

Actual / Predicted	Predicted: No (0)	Predicted: Yes (1)
Actual: No (0)	True Negative TN	False positive (FP)
Actual: Yes (1)	False Negative FS	True positive (TP)

Objective:

To evaluate the classification performance of the Logistic Regression model.

b) Interpret the results.

Interpret each component of the confusion matrix:

- **True Positives (TP):** Loan applications correctly predicted as approved.
- **True Negatives (TN):** Loan applications correctly predicted as not approved.
- **False Positives (FP):** Loan applications incorrectly predicted as approved.
- **False Negatives (FN):** Loan applications incorrectly predicted as not approved.

Based on these values, discuss whether the model performs well in identifying approved and rejected loan applications.

Q10. Model Evaluation – ROC Curve & AUC

a) Plot the ROC curve.

b) Calculate the AUC score.

SOL : a) Plot the ROC Curve.

Generate a **Receiver Operating Characteristic (ROC) Curve** using the predicted probabilities obtained from the Logistic Regression model.

The ROC curve illustrates the trade-off between:

- **True Positive Rate (Sensitivity or Recall)**
- **False Positive Rate (1 – Specificity)**

The curve helps assess the model's classification performance across different decision thresholds.

Objective:

To visualize the discriminative ability of the Logistic Regression model.

b) Calculate the AUC Score.

Compute the **Area Under the ROC Curve (AUC)** value.

The AUC score ranges from **0 to 1**, where higher values indicate better model performance.

General interpretation guidelines:

AUC SCORE	INTERPRETATION
0.50	No discriminative ability
0.60-0.70	poor performance
0.70-0.80	Acceptable performance
0.80-0.90	Excellent Performance
Above 0.90	Outstanding performance

Objective:

To quantify the model's ability to distinguish between approved and non-approved loan applications.